

Chapter 1: Social Influence and its Translation to the Legislative Arena

1.1 Introduction

The expanded role of technology in American politics has changed how legislators connect and share information with other legislators, constituents, and the mass public. Rep. Alexandria Ocasio-Cortez (D-NY-14) has over 12.7 million followers on Twitter,¹ her reach far extending beyond the 725,000+ people in her district. Legislators use these digital platforms to relay both formal information to their constituents, such as news regarding bills they are working on or their campaign, and informal information, such as pictures of their dogs and children. Legislators also use digital platforms to interact with each other, encouraging support for specific bills and sharing the successes of their working groups within Congress. However, little is understood about how digital technologies are changing the roles, strategic connections, and mechanisms for legislative coalitions or how accurately Twitter is reflecting legislative influence and effectiveness.

If we consider the number of followers one has, which is the most basic measure of influence on social media, it could be the case that those with more followers are the more effective legislators. However, it could also be the case that a different measure which captures legislators relationships with each other drives legislative effectiveness. The difference is important because if basic public facing measures do not lead to increases

¹Twitter was re-branded to X on July 23rd, 2023.

in legislative effectiveness but a different measure does and the public is not receiving information about this alternative measure this could give the public a skewed sense of the influence different legislators have. Additionally, the literature has thus far studied relationships between legislators offline and their impact on legislative output. This chapter will explore what online relationships between legislators on social media can tell us about legislative output. It could be the case the more informal legislative network via Twitter does not inform us about legislative output, suggesting social media is separate from legislating. However, it could also be the case that these connections can tell us about legislative output indicating social media influence the legislative process. If connections on social media between legislators can inform us about legislative output this is important because it suggests that we can glean important information regarding the relationships between legislators from Twitter, allowing us new avenues into exploring legislative connections, coalitions and success.

There are multiple goals for this chapter, the first of which is to discover what makes a legislator effective in the 21st century. Namely, considering what role, if any, Twitter plays in representing the effectiveness of a member of Congress. I first consider a modern-day measure of political influence the public knows how to interpret, number of followers. I hypothesize this measure will have no effect on legislative effectiveness, as measured by Volden and Wiseman (2009). The results are consistent with this hypothesis and indicate the number of followers a legislator has does not lead to increased legislative success.

Given this, I aim to discover what Twitter can tell us about legislative output. To do this, I consider what legislator interactions on social media can tell us about the more formal legislative environment. I follow a simple intuition: that connections between legislators are suggestive of an underlying social or professional relationship. Specifically, I consider the network of interactions between legislators on Twitter, via retweets and mentions, and the network of interactions legislators have within Congress, via cosponsorships, and provide thorough descriptive information.

I then take this network of interactions and transform it into a measure of connectedness (Fowler, 2006), or how embedded a legislator is within each network legislative and compare and compare and contrast leaders in both networks, highlighting their similarities and differences. I find those who are the most connected within Twitter are not those who are the most connected within the more formal cosponsorship network. This finding suggests relationships between legislators on Twitter are capturing something distinct from the relationships within the cosponsorship network.

Fowler (2006) demonstrated the more connected a congressperson is within the cosponsorship network, the more effective they are as a legislator. I expect regardless of contextual environment, those who are most embedded socially within either network will be the most effective. This transforms the simple notion of a cosponsorship or retweet from being a small signal of a social relationship between legislators to say, in the aggregate, if we consider the frequency, strength, and quantity of connections between legislators we can infer legislative effectiveness through connectedness. A more connected legislator is likely to have increased legislative success because they have stable, in terms of quantity and quality, connections to pull on in the network to advance legislation.

To test the relationship between connectedness and legislative effectiveness, multiple regressions are run. I find connectedness within the cosponsorship network is positively and significantly related to legislative effectiveness, consistent with past findings. I then also show connectedness within the Twitter network is positively and significantly related to legislative effectiveness, i.e. the more connected one is the more effective they are, even when controlling for seniority as well as a multitude of other common controls. This indicates Twitter reflects the underlying social structure of Congress through retweets and mentions and that those ties' frequency and strength are indicative of legislative success.

Taken together, I argue my findings provide evidence Twitter accurately reflects the social relationships between members of Congress and that those social relations are tied to legislative success. However, there is a disconnect between what individuals see on Twitter and interpret as influence (number of followers) and what is going on behind

the Capitol walls. Specifically, increases in the number of followers legislators' accounts have do not lead to increased effectiveness. Instead, it is the, somewhat obscure, quantity connectedness that results in increased effectiveness for legislators. Even though on the surface a retweet may seem inconsequential, my findings indicate otherwise— that they are demonstrative of true social and professional relationships between legislators.

In sum, it is not those with the most followers who are the most effective legislators, but rather those who are the most connected; this is contrary to preconceived notions of social influence (Cha et al., 2010). Additionally, connections between legislators on social media can provide insight into working relationships within Congress. More research can and should be done using information from these connections. Possible areas for future work include exploring the impact of the dissonance between publicly perceived influence and congressional influence, as I expect this skewed sense of influence can have negative implications for Twitter users' satisfaction with the government, as they are not seeing influence accurately represented on Twitter. Lastly, this paper paves the way for more studies regarding interactions between legislators on social media platforms and what we can glean from those relationships, as the underlying network of legislative interactions is basically untapped.²

The goal of this paper is thus threefold. I first will explore if social influence, via number of followers, is synonymous with legislative influence. Then, I will confirm Fowler's findings for the 115th and 116th Congresses, demonstrating connectedness within the cosponsorship network indicates legislative success. Lastly, I will extend the analysis to consider legislators' interactions with each other on Twitter and determine if connectedness on Twitter suggests legislative effectiveness.³

²One notable exception being Peng et al. (2016) which demonstrated legislators who follow each other on Twitter are more likely to have increased vote agreement on roll call votes, highlighting the impact social media connections can have on the legislative process.

³Data and code needed to evaluate the conclusions of the paper are available on GitHub at <https://github.com/caramnix/social-influence>.

1.2 Legislating in the 21st Century

1.2.1 Modern Day Connections

Before we can begin analysis, we must first consider how to determine connections between legislators, as this is a key concept/notion in our analysis. In the literature on social networks, a connection is usually defined as some sort of relationship between individuals. This can be a friendship, communication via email or online messaging platforms, membership in a common group or organization; anything that occurs between two (or more) people.

The importance of networks for the policy process has been explored through multiple different avenues. Through Dear Colleague letters (Craig, 2021; Box-Steffensmeier, Christenson and Craig, 2019), lobbying networks (Box-Steffensmeier and Christenson, 2014; Box-Steffensmeier, Christenson and Hitt, 2013; LaPira, Thomas and Baumgartner, 2009), and legislative networks (Kirkland, 2011, 2013; Kirkland and Gross, 2014).

On Twitter, there are clear ways legislators are interacting with one another, they follow each other, retweet/quote tweet each other, and tag each other in tweets. However, legislation is still created within the constraints of House and Senate rules and any product of the House or Senate will be subject to a level of legislative formality. Thus, it becomes more difficult to glean relationships between members. Past scholarship has considered college attendance (Cohen and Malloy, 2014), committee assignment, state kinship, shared partisanship (Caldeira and Patterson, 1987), seating within Congress (Masket, 2008) to try and extract relationships from this inherently formal environment, finding these relationships impact voting behavior and can impact electoral collaboration as well (Box-Steffensmeier et al., 2020). I will utilize the network of legislative cosponsorships which plays a significant role in the legislative process. Cosponsorships are a form of communication- and can signal shared interests, beliefs, and friendships. The House rules as they currently stand allow only one legislator to be listed as the sponsor of legislation, regardless of if the legislation was a joint effort by a group of legislators. Therefore, past

scholars have dismissed the importance of legislative cosponsorship due to them being a weak signal of support (Wilson and Young, 1997). Following Fowler (2006) I argue that though it may signal weak support for the specific legislation, it provides a strong measure of social connection within the legislative body and these connections can provide us insight into legislative output.

In sum, we consider two different forms of social connections, one for each contextual environment: (1) legislative cosponsorship and (2) retweeting or mentioning another legislator on Twitter.

1.2.2 How are Legislators Using Twitter?

Surprisingly, even though legislators interact with each other on Twitter frequently, scholars have devoted little attention to these interactions, focusing instead on the relationships between the public and the legislators (Barberá et al., 2019). How legislators interact with the public on social media is important, as social media allows them direct access to communicate with their constituents. However, Twitter also allows legislators more access to their colleagues, so they are more easily able to communicate with and promote each other's work on these platforms. Twitter also allows legislators to reach a vast audience very quickly, as opposed to say, putting out a press release in a constituent email or on their website.

Prior research has shown Twitter is not being used by legislators to find support from congresspeople on Twitter for legislation, but rather to disperse information (Golbeck, Grimes and Rogers, 2010), as legislators tend to have one-way conversations with each other on Twitter (Mergel, 2012). It is not a discussion but rather a declaration- either declaring, I worked with @SenatorX, or I disagree with @CongresspersonY. Rarely, if ever, is there a back-and-forth discussion on Twitter between congresspeople leading to consensus and new legislation. This mirrors the role of the cosponsorship network; it is not used to build coalitions, but rather reflect relationships that already exist.

The majority of retweets and mentions between legislators then reflect social and

professional connections with people they already know will help them advance legislation. Twitter's main use for congresspeople is self-promotion (Golbeck, Grimes and Rogers, 2010), with 53% of tweets being informational. Figure 1.1 provides an example of this use of Twitter, where Sen. Klobuchar (D-MN) self-promotes her work on the Equality Act, but does not tag or mention by name any other legislator.

Research from 113th Congress also finds the largest category of tweets is professional (Mergel, 2012). 27% of tweets were found to pertain to legislator's location or activity, providing a more social feel where constituents feel included in their daily lives and activities.

However, this paper is focused on the connections between legislators on social media. Social media posts from legislators frequently reference other legislators they are working with. For example, Figure 1.2 is a tweet from Sen. Klobuchar where she mentioned, and tagged, Sen. Wyden (D-OR) as someone she was introducing a bill with—signaling a working relationship between the two.

Retweets, similar to tweets, can also signal both formal and informal connections between legislators. For example, Figure 1.3 shows a tweet where Rep. Ocasio-Cortez retweeted Rep. Pressley's tweet discussing the introduction of a bill. Rep. Ocasio-Cortez also tags other members of "the Squad" in the tweet.

Another example can be seen in Figure 1.4, which is a retweet from Rep. Alexandria Ocasio-Cortez where she retweets Rep. Jahana Hayes (D-CT) which signals a friendship or working relationship between the two legislators, one which is not as obvious as Rep. Ocasio-Cortez's ties to the "Squad."

Importantly, retweets are used to not only boost tweets which mention the legislator themselves, but also to boost the accomplishments of others within Congress. The expectation is retweets will also indicate a social connection and give us valuable insight into not only into the professional working relationships of congresspeople, but also the social relationships between them. This can be seen in Figure 1.5, as Rep. Ocasio-Cortez retweeted an article discussing Rep. Jamayal (D-WA-7) and tagged her to show support

for her friend within Congress.

Clearly, retweets and mentions between legislators do seem to signal friendships and working relationships, making them an excellent source of data into legislative friendships and working relationships.

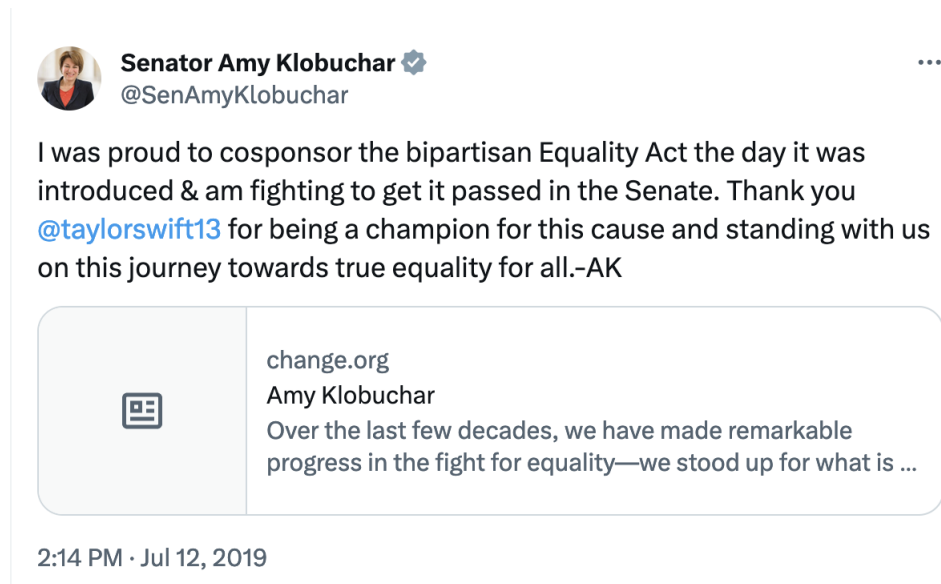


Figure 1.1: Tweet from Sen. Klobuchar on 7/12/2019.

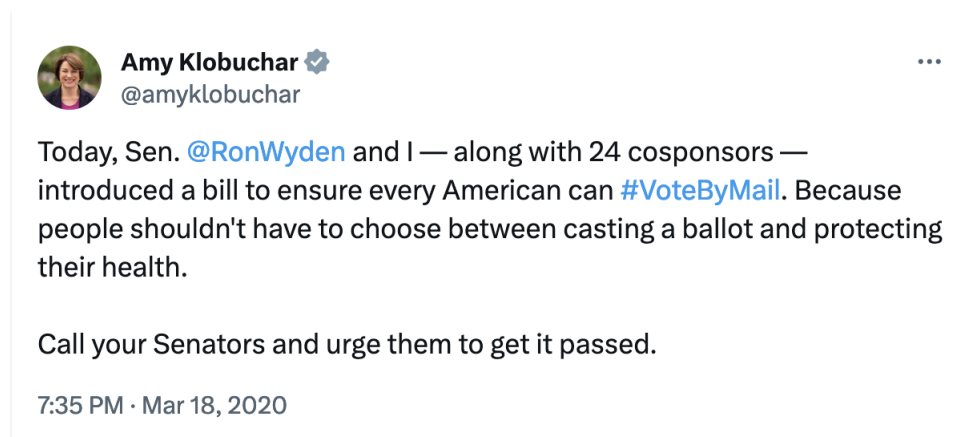


Figure 1.2: Tweet from Sen. Klobuchar on 3/18/2020.



Figure 1.3: Retweet from 1/4/2019 between Rep. Pressley and Rep. Ocasio-Cortez.



Figure 1.4: Retweet from 3/31/2019 between Rep. Jahana Hayes and Rep. Ocasio-Cortez.



Figure 1.5: Rep. Ocasio-Cortez's Retweet from 3/22/2019.

1.3 Theory and Hypotheses

Rep. Alexandria Ocasio-Cortez quickly became a social media powerhouse following her election into the 116th Congress. She currently has amassed over 12 million followers⁴ and uses her platform to boost legislation she is working on as well as the legislation of her close colleagues. While this influence on social media is impressive one might wonder how this power translates into the legislative arena. Using legislative effectiveness scores from Volden and Wiseman (2009), one would see Rep. Ocasio-Cortez was ranked 230/240 of the Democrats in the 116th Congress. This score indicates at first glance we should not expect legislators who are powerful on Twitter to necessarily be powerful at progressing legislation within the Capitol walls. Put plainly- legislative effectiveness is completely separate from the number of followers one has.

However, in the 21st century, the number of followers an individual has on social media indicates a level of influence within their community. When a Twitter user engages with a legislator on Twitter, they see the number of followers a legislator has. Given this is one of the only measures of social influence available to users it is not difficult to imagine they might incorrectly directly associate social influence with legislative influence. As researchers, we might immediately find flaws in this association, namely that the ideological composition of Twitter is biased, with more liberals on the platform than conservatives (Wojcik and Hughes, 2019), the average age of legislators on Twitter is younger than the average age of a congressperson, and we know that viral tweets do not easily become legislation.

For a specific example, let's revisit Rep. Ocasio-Cortez. She is known for her "radical" left agenda and was involved in introducing the Green New Deal into the House. Her tweet introducing the Green New Deal gained 4,000+ retweets and 29,000 + likes, which is viral for a legislator. An average Twitter user might misperceive her influence within Congress due to the support she has via retweets and likes, as well as because of the number of followers she has. Therefore, I hypothesize that,

⁴As of February 2021.

H1: *There is not a relationship between the number of followers a legislator has on Twitter and legislative effectiveness.*

This hypothesis has important normative implications, namely, if true, it implies Twitter is incorrectly portraying the working structure of Congress. With over half of Americans getting their news “sometimes” or “often” from social media (Shearer, 2021) it is important Twitter accurately reflects legislative power. It is possible this disconnect could increase frustration with the government and increase animosity toward the out-party. Prior research has shown most Americans do not understand the legislative process (Achen and Bartels, 2016), nor can they discriminate between effective and ineffective lawmakers. However, there is a relationship between one’s perception of effectiveness and the approval of the legislator (Butler et al., 2021). So, if Twitter users conflate social influence with effectiveness, they may in turn approve of legislators who are in reality not effective, thus supporting legislators who are not getting things done for them which could in turn cause frustration with the government.

However, if the number of followers a legislative has on Twitter is not indicative of legislative success, what if anything, can Twitter tell us about legislative success? To explore this, I consider the work of Fowler (2006) where he showed social connections between congresspeople can predict legislative effectiveness. Specifically, he showed from 1973 to 2004 that the more connected a congressperson is, the more effective they are. But can this idea be extended to more recent Congresses and social media connections? I argue one’s embeddedness in the network, or how connected they are, will be related to their legislative effectiveness in both contextual environments.

As Fowler demonstrated for the congressional cosponsorship network in 2006, how connected someone is within the cosponsorship network represents a social connection or friendship giving a legislator the ability to pull on some strings (edges) to get support for legislation. Legislating is a cooperative activity. If one wants to advance their legislative agenda, they need the help of their colleagues to do so. Therefore, it is not hard to imagine

legislative success is determined based on one's ability to work with other legislators, both of the same party and across the aisle. In fact, research has shown it is the weak ties between legislators of opposite parties that are the most vital for increasing legislative success (Kirkland, 2011). Legislative success is inherently tied to how embedded someone is within the network, how many friends one has, and how many favors one can call in to pass legislation. Connectedness is able to capture the intricacies of the relationships within the network, considering the connections of connections. This makes it a reasonable measure to utilize.

Members who are better connected within the congressional network have relationships with more of their colleagues, particularly more close relationships, where the frequency and intensity of support is greater (i.e. they cosponsor legislation for someone a lot and are frequently one of only a few people to do so). Congress acts as a social network (Porter et al., 2005) and the power of that social network should not be discounted, as connections between legislators have been shown to influence floor voting behaviors (Masket, 2008; Cohen and Malloy, 2014) as well as increase the share of federal grant money a legislator obtains (Craig, 2021). Therefore, I expect to find support for Fowler's findings regarding the cosponsorship network for more recent Congresses.

H2: *The more a legislator is embedded/influential within the legislative cosponsorship network the more effective they are as a legislator.*

Social connections can also be found between legislators' interactions within different environments. This paper will focus on the Twitter environment. I believe an underlying social network can be created from retweets and mentions on Twitter and these edges will represent both social and working relationships. Let's consider another example, Sen. Amy Klobuchar (D-MN) has 1.8 million followers on Twitter and was the seventh most effective Senator in the 116th Congress. She is fairly influential on Twitter and an extremely effective legislator.

As shown above, we know Rep. Ocasio-Cortez and Sen. Klobuchar both use Twitter to convey information to the public on bills they are working on and retweet to support content that agrees with their brand, and yet both of these legislators vary widely in terms of legislative success. So, then what is it that differentiates these two legislators? It is not difficult to imagine Rep. Ocasio-Cortez and Sen. Klobuchar have different types of connections and friendships within Congress, in part due to seniority. Legislation takes a village; it is inherently cooperative as one needs the support of other congresspeople to advance legislation

Therefore, I believe Twitter can inform us about legislative effectiveness through the network of legislative interactions, i.e. those who are most influential or embedded within the Twitter network will also be the most effective.

I expect to find that even though Twitter is an informal place for legislators to communicate with each other and their constituents it captures valuable social and professional connections between legislators which can infer legislative success. As I showed in the section above, legislators will retweet each other and mention each other in tweets related to legislation they are working on together, and also more informally, to show support for each other's work separate from cosponsorship. Therefore, I expect,

H3: *The more a legislator is embedded/influential within the Twitter network the more effective they are as a legislator.*

1.4 Data

To assess the hypotheses laid out in the previous section I (1) calculate the connectedness within each contextual environment and (2) join this data with legislative effectiveness information. I was able to rely on the definition of connectedness laid out in Fowler (2006) for the cosponsorship network. I was also able to apply a similar routine to calculate the connectedness within the Twitter network. Before I discuss this

measurement, I first discuss the congressional bill data and Twitter data I utilize to calculate this measure.

1.4.1 Cosponsorship

The cosponsorship network is defined by legislative cosponsorship in the U.S. House and U.S. Senate for the 115th and 116th Congress (Jan 3, 2017- Jan 3, 2021). Importantly, the rules on cosponsorship did not change during this period. I define legislation to consist of all House Bills (H.R., H. Res.) and Senate Bills (S., S. Res.), which constitute all formal legislative proposals before the House and Senate during the 115th and 116th Congresses. For simplicity, and following previous literature, I treat cosponsorship and sponsorship as equal. This decision is also due to the fact congressional rules state there can only be one sponsor per bill, and yet legislation in its entirety is rarely crafted by just one legislator- it is a joint effort within the legislative body.

The resulting network is a one-mode (“unipartite”) network where a member of Congress is connected to another by an edge if one has cosponsored another. The network is also weighted, in order to take into account the frequency in which legislators cosponsor one another. This is encoded using a unipartite adjacency matrix M , with entries weighted entries indexed as w_{ij} , where w_{ij} is equal to the number of times legislator i cosponsored legislator j and 0 if there were no cosponsorships between them. Importantly, cosponsorships are directional, so the resulting matrix is not symmetric.

The cosponsorship matrix from the 116th House contains information on the cosponsorships of over 10,300 bills introduced in the House and contains 172,580 cosponsorship ties. This adjacency matrix is then transformed into a weighted directed network object, where an edge between legislator i and legislator j corresponds to the frequency of cosponsorships between i and j , see Figure 1.6 for a visual representation of this network. 14 percent of cosponsorships were cross-party. Inter-party cosponsorships were dominant, with 66 percent of cosponsorships being between Democrats and 16 percent between Republicans, this is intuitive as the Democrats were in control of the House at the time,

so they would have had the opportunity to cosponsor each other more frequently.

The density of this network object is .34. On average each legislator in the 116th Congress cosponsored 385 pieces of legislation (or 3.7 percent of all possible legislation).

Please see Table 1.1 for the descriptive information for all sessions and chambers of Congress considered in this study.

Network of Cosponsorships, 116th House

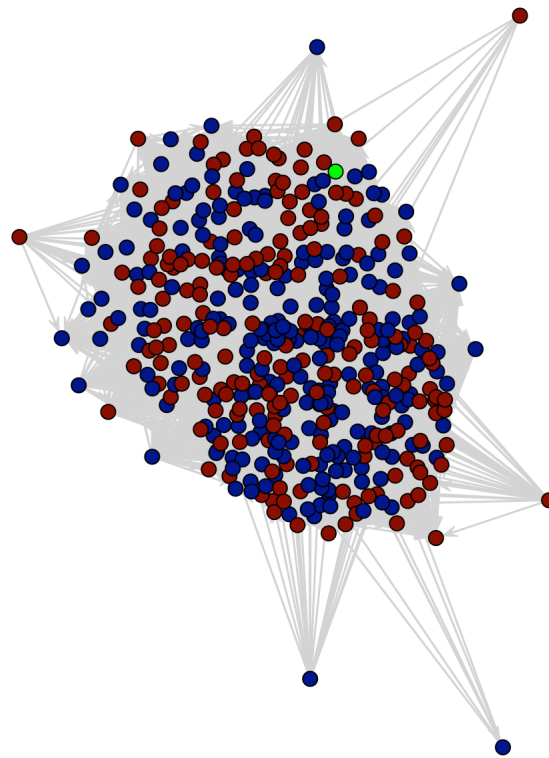


Figure 1.6: 116th House cosponsorship network, node color indicates political party.

1.4.2 Twitter

The Twitter network is defined by legislator retweets and mentions on Twitter for members of the U.S. Congress for the duration of the 115th and 116th Congresses, with analysis done separately for each Congress and each chamber. It consists of all interactions

Congress	Bill Quantity	Cosponsorship Ties	Average Cosponsorships
116th House	10,335	172,580	385
116th Senate	5,894	39,170	384
115th House	8,581	133,079	294
115th Senate	4,552	28,621	273

Table 1.1: Descriptive statistics for cosponsorship networks.

between legislators on Twitter, with retweets and mentions treated as equivalent. The data was collected using the *academictwitteR* package (Barrie and Ho, 2021) using the *Twitter API for Academic Research* (2021). The goal of looking at retweets and mentions on Twitter is to isolate both personal and professional relationships. Exploration of legislators’ Twitter feeds was informational as it confirmed most legislators tweet about legislation they are working on with other members of Congress. They are also likely to retweet the accomplishments of friends or colleagues. Therefore, both retweets and mentions are vital for considering the relationships between legislators, as shown and discussed previously.

For the 116th House there were 908,820 tweets by legislators during the duration of the 116th Congress, of those 145,183 mentioned other legislators (16 percent). A weighted directed network object is created from the adjacency matrix. The resulting network is thus a one-mode (“unipartite”) network where a member of the House of Representatives is connected to another by an edge if one retweeted or mentioned another in a tweet. The network is of course weighted to reflect the frequency with which legislators retweeted or mentioned one another. This is encoded exactly as above using a unipartite adjacency matrix M , with entries M_{ij} equal to w_{ij} , which corresponds to the number of times legislator i retweeted or mentioned legislator j then and 0 if legislator i never retweeted nor mentioned legislator j . See Figure 1.7 for a visualization of the Twitter network. ⁵

Inter-party retweets were dominant, with 66 percent of retweets being between Democrats and 24 percent between Republicans. Notably, despite Republicans and

⁵For an interactive version of this figure, please see [here](#).

Network of Twitter Interactions, 116th House

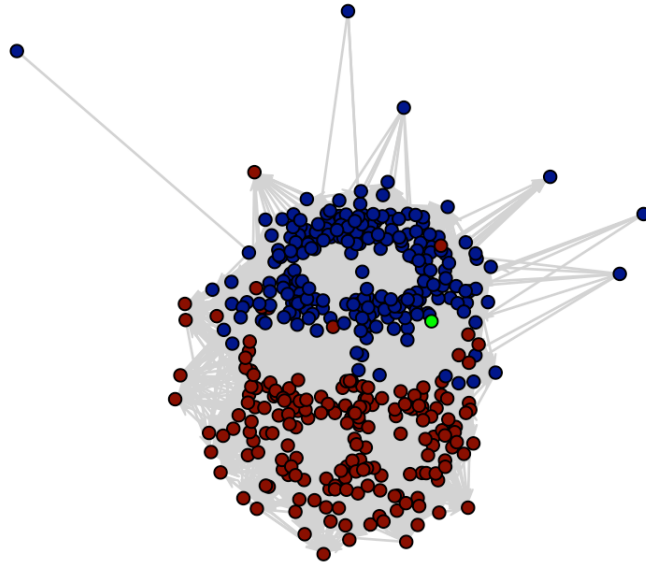


Figure 1.7: 116th House Twitter network, node color indicates political party.

Democrats having the same access to Twitter, it is clear Republicans are using Twitter less than their colleagues across the aisle. Figure 1.7 shows a clear divide between Democrat and Republican nodes. This is because only 14 percent of tweets were cross-party, a decrease of 5 percent from the cosponsorship network. This indicates less intra-party collaboration and communication on Twitter than is reflected within the cosponsorship network; Congress is legislating together more than they are tweeting together. Therefore, Twitter is reflecting a more polarized and segmented Congress than reality, which does have implications for public perceptions of polarization, as it perpetuates the narrative that legislators are not working together.

The descriptive statistics for the rest of the networks can be found in Tables 1.2 and 1.3.

Congress	Num. of Tweets	Num. which Mention Another Legislator(s)
116th House	908,820	145,183
116th Senate	296,485	36,889
115th House	675,492	106,083
115th Senate	219,903	29,368

Table 1.2: Descriptive statistics for Twitter networks.

Congress	Num. of Tweets	Num. Retweets/Mentions	Num. Without Self Ties	Average Num. Tweets
116th House	908,820	196,355	130,645	298
116th Senate	296,485	50,924	30,626	300
115th House	675,492	134,453	72,642	160
115th Senate	219,903	78,894	44,387	396

Table 1.3: Descriptive statistics for Twitter networks.

1.4.3 Legislative Effectiveness

The last component of my analysis is legislative effectiveness. The most widely used measure of legislative effectiveness is legislative effectiveness scores (LES scores) created by Volden and Wiseman (2009). The legislative effectiveness scores for all members of the 115th and 116th Congresses were collected.

What goes into these scores? LES scores incorporate fifteen indicators of legislative success. The first step is to categorize all bills into three categories: commemorative (C), substantive (S), or substantive and significant (SS). The weight of importance for commemorative bills is less than that of substantive, which is less than that of substantive and significant. So, the most effective legislators at this stage would sponsor only substantive and significant bills. Within these three categories there are five classifications for how far a bill progressed through Congress, namely the number of bills that member of the House of Representatives sponsored (BILL), how many of those received any action in committee (AIC), or action beyond committee (ABC) on the floor of the House.

For those bills that received any action beyond committee, it is also considered how many of those bills subsequently passed the House (PASS), and how many became law (LAW). Therefore, building on the previous statement of effectiveness, the most effective legislator would introduce only substantive and significant bills that became law.

For example, if we consider Rep. Ocasio-Cortez, she was ranked 230/240 for legislative effectiveness among Democrats for the 116th Congress. This is because she introduced 21 bills, all of which were substantive (S), but none saw any action outside of being introduced.

Additionally, Volden and Wiseman (2009) argue legislative effectiveness is a function of innate ability, cultivated skills, and institutional positions, as such they provide variables related to legislators' observed characteristics which are theoretically important for legislative effectiveness and thus should be controlled for. First, the length of time a legislator has been in Congress. As legislators spend more time in Congress they are expected to become better and more effective lawmakers.

The second factor is the size of their congressional delegation which counts the number of congresspeople from the state they are representing. Larger congressional delegations may make it easier for legislators to obtain support for their legislation. The third factor is the vote share they obtained to enter the most recent congress, as those who received more support may not feel as pressured to produce legislation. The next two factors pertain to ideology and partisanship. The first dimension of DW-Nominate (McCarty, Poole and Rosenthal, 1997) is included as well as a control for political party identification. Individuals with more moderate ideologies may have an easier time finding support for their legislation within Congress.

Four measures that relate to the institutional arrangement within Congress are also considered, if a member is a committee chair, majority or minority leader, and, for the House analysis, if they are Speaker. Theoretically, positions of power within are important for legislative effectiveness as it may allow them to garner support more easily for legislation within their party, making them more effective legislators. Their leadership status also impacts the raw number of bills they are able to introduce, as House majority leaders, committee leaders, and subcommittee leaders introduce more bills than the average house member giving them more chances for legislative success (Anderson, Box-Steffensmeier and Sinclair-Chapman, 2003). Additionally, those in positions of power

may already be uniquely effective legislators to have obtained those positions in the first place.

Lastly, controls for gender as well as if a legislator is Black or Latinx are also included following literature from Volden, Wiseman and Wittmer (2013), Rocca and Sanchez (2008), and Bratton and Haynie (1999).

1.5 How Influential Are You?

1.5.1 Number of Followers

Recall I want to consider a public-facing measure of social influence. I use the number of followers a legislator has as a public-facing proxy for social influence as this value is visible and easily interpretable by the public. Due to data constraints, I will only be able to consider this relationship for the 116th House. The data on the number of followers was collected from Twitter in February of 2021.

The top five legislators with the most followers at the time of collection are displayed in Table 1.4. As expected, Rep. Ocasio-Cortez tops the list. Then Rep. Pelosi (D-CA-12), who was Speaker of the House at the time, followed by Rep. Omar (D-MN-5), a member of the Squad, and then Rep. Schiff (D-CA-28), the ranking Democrat for the House Select Intelligence Committee at the time and then Rep. Jordan (R-OH-4), the ranking Republican for the House Oversight and Government Reform committee at the time. These are all legislators who we'd expect to have large followings for one reason or another.

Legislator	Number of Followers
Rep. Ocasio-Cortez, Alexandria [D-NY-14]	12,544,125
Rep. Pelosi, Nancy [D-CA-12]	7,016,986
Rep. Omar, Ilhan [D-MN-5]	2,965,967
Rep. Schiff, Adam B. [D-CA-28]	2,928,465
Rep. Jordan, Jim [R-OH-4]	1,913,870

Table 1.4: Top five legislators with the most Twitter followers, 116th House.

1.5.2 Connectedness

Connectedness is a measure created specifically for the cosponsorship network by Fowler (2006). It aimed to measure the social connections between legislators, and it was shown to have a positive relationship with legislative influence, which was defined by the number of floor amendments a legislator was able to pass. It takes into account two pieces of information unique to cosponsorships that are not considered by other measures of centrality: (1) information regarding the total number of cosponsors on each bill and (2) information about the total number of bills sponsored by j that are cosponsored by i . The main idea is bills with fewer total cosponsors provide more reliable information about the real social connections between two legislators than bills with many cosponsors (Burkett, 1997) and legislators who frequently cosponsor each other's bills have stronger connections than legislators who only sponsor a few of each other's bills. Specifically, four steps go into calculating connectedness:

1. For each bill, b one assigns a weight value of $1/\text{total number of cosponsors on bill } b$ from each legislator who cosponsored the bill to the bill's sponsor. The logic is that the fewer number of cosponsors the more significant the relationship between legislator i and j was. The result is an $n \times n$ matrix w_{ij} which reflects the weighted quantity of cosponsored bills, where n is the number of legislators. In this matrix, bigger values imply stronger connections.
2. For each entry in w_{ij} , take $\frac{1}{w_{ij}}$. The resulting matrix, d_{ij} , is an $n \times n$ matrix and bigger values imply less connected legislators.
3. Dijkstra's algorithm (Dijkstra, 1959) is then run on this matrix. The result is an $n \times n$ matrix, $dijkstra_{ij}$, of shortest distances from each legislator to every other legislator. In this matrix, bigger values imply weaker connections between legislators.
4. Lastly, in order to obtain one singular value for the connectedness of each legislator take $\frac{n-1}{\sum_{i=0}^n dijkstras_{ij}}$. In this matrix, bigger values imply stronger connections between legislators, so the legislator with the largest value is deemed the "best connected."

Connectedness was created to apply to cosponsorship networks. Therefore, in order to consider the connectedness of legislators on Twitter I extended the measure to be used on social media connections. This was done by using the four main steps laid out above with one big difference: instead of considering the cosponsorships on individual bills I

instead considered the mentions within each individual tweet. Specifically, for each tweet by a legislator that mentioned any number of legislators, n . The weight added to each person mentioned in the tweet by the author is $1/n$, just as above.

As tweets are limited in the number of characters they can have this also limits the number of legislators one can mention. However, I believe this to be a benefit of using interactions between legislators on Twitter as only the legislators closest to each other will be mentioned and only those most integral to the bill will be mentioned. This will also get rid of some of the noise found within the ties of the cosponsorship network. For example, when a legislator tweets regarding their proposed bill, it is likely they will tag only those legislators who helped with the substantive content of the bill. This contrasts with using the cosponsorship network, as there is no way to distinguish from all of the bill's cosponsors who actively contributed to the bill's content. The same logic outlined above is applicable for Twitter; tweets that mention only one legislator, as opposed to three or four provide more reliable information about the social and professional connections between two legislators. Similarly, legislators who retweet or mention tweets written by the same legislator are more likely to have a close social relationship than those who only retweeted or mentioned a legislator a few times. Thus, the 280-character limit will only change the strength of the connections between legislators and the analyses will still be able to be run and interpreted in the exact same way and connectedness can be theoretically extended to the Twitter network.

As a test of the connectedness measure on Twitter consider the "Squad," a group of four freshman congresswomen during the 116th Congress who were known for their strong social media and legislative friendship. The original Squad's members were Rep. Alexandria Ocasio-Cortez (D-NY-14), Rep. Ilhan Omar (D-MN-5), Rep. Ayanna Pressley (D-MA-7), and Rep. Rashida Tlaib (D-MI-13). When considering the top-5 connections, based on connectedness, for each member of the Squad the three other members of the Squad were present. In sum, they were all shown to be strongly connected to each other, see Table 1.5 for Rep. Ocasio-Cortez's top five connections via connectedness and see

Figure 1.8 for a corresponding figure containing the top five connections for each Squad member. Each Squad member is connected to the other three Squad members. The grey-colored nodes are non-squad members who are connected to a Squad member. As one can see, there is overlap between these members as well, suggesting a tight-knit Twitter community. For example, Rep. Omar and Rep. Tlaib both have Rep. Pocan (D-WI-2) in their top five connections.

Similarly, for the 116th Senate, the connectedness measure identifies the “bromance” found between Senators Booker (D-NJ), Schatz (D-HI), and Murphy (D-CT). For each of the three Senators, the other two are in their top-5 connections. This further provides evidence that Fowler’s connectedness measure was successfully extended to the Twitter contextual environment.

Rep. Ocasio-Cortez’s top five connections.
Rep. Tlaib, Rashida [D-MI-13]
Rep. Omar, Ilhan [D-MN-5]
Rep. Pressley, Ayanna [D-MA-7]
Rep. Long, Billy [R-MO-7]
Rep. Maloney, Carolyn B. [D-NY-12]

Table 1.5: Rep. Ocasio-Cortez’s top five connections

The top five most connected congresspeople for the 116th House in both contextual environments can be seen in Tables 1.6 and 1.7. Notably, there is no overlap between these two tables. This indicates being connected on Twitter does not imply a legislator is connected within the cosponsorship network. For the top five most connected for the other sessions and chambers of Congress please see Appendix A.

It is also worth noting there is overlap between those with the most followers and those who are most connected within the Twitter network. Specifically, Rep. Pelosi, Rep. Schiff, and Rep. Ocasio-Cortez are present in the top five for both measures of influence. And yet, the correlation between them is 0.208. This suggests our original inference is

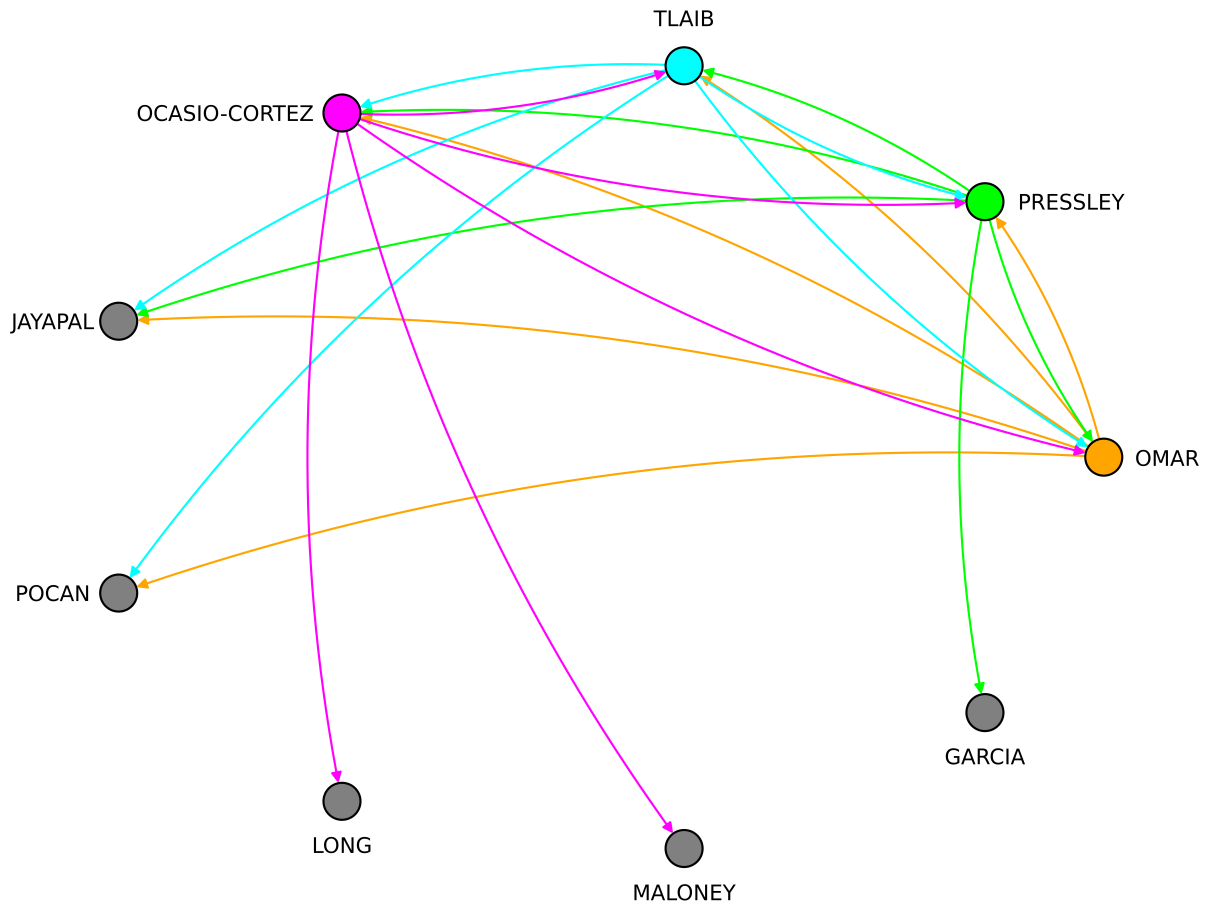


Figure 1.8: Figure containing top five connections for each Squad member. The color of each arrow corresponds to the legislator from which each connection is coming from.

correct: these measures capture different things.

Notably, the values in the table above are the raw scores obtained via the connectedness measure. For all analyses done below connectedness is scaled to be between zero and one.

Legislator	Connectedness
Rep. Pelosi, Nancy [D-CA-12]	2.95
Rep. Schiff, Adam B. [D-CA-28]	2.74
Rep. Ocasio-Cortez, Alexandria [D-NY-14]	2.58
Rep. Nadler, Jerrold [D-NY-10]	2.56
Rep. Dingell, Debbie [D-MI-12]	2.55

Table 1.6: Connectedness: top five legislators within the 116th House Twitter network.

Legislator	Connectedness
Rep. Norton, Eleanor Holmes [D-DC-At Large]	0.244
Rep. Raskin, Jamie [D-MD-8]	0.235
Rep. Fitzpatrick, Brian K. [R-PA-1]	0.226
Rep. Rush, Bobby L. [D-IL-1]	0.225
Rep. Beyer, Donald S., Jr. [D-VA-8]	0.223

Table 1.7: Connectedness: top five legislators within the 116th House cosponsorship network.

1.6 Results

In this section, I present the results of multiple regression analyses which assess the hypotheses I laid out earlier in the paper. Recall this paper had three main objectives. First, to determine if social influence, measured by number of followers a legislator has on Twitter, is synonymous with legislative influence. I expected this would not hold we must use interactions between legislators to determine legislative success instead. Second, to validate Fowler’s findings for more recent Congresses, i.e. demonstrate cosponsorship connectedness is associated with an increased legislative influence, as measured by LES scores rather than Fowler’s original measure using floor amendments. I expect to find Fowler’s results will hold for more recent Congresses.

My third and final hypothesis is that the relationship between social connections and legislative effectiveness will also hold within the Twitter contextual environment. To assess all three of these hypotheses I ran multiple linear and fixed-effects models. By selecting this modeling approach, I was also able to control for an obvious indicator of legislative success, seniority, measured by the number of years one has been in Congress

as well as other common indicators of legislative success.

Let's assess my first hypothesis, if we measure social influence using the most basic measure available to the public, via number of followers, there is not a statistically significant relationship between number of followers and legislative effectiveness, as expected and laid out in hypothesis 1. The results are shown in Table 1.8 for the 116th House. As one can see, there is minimal effect between the number of followers legislators have and legislative effectiveness when controlling for seniority and other common controls. This implies the public may be misattributing legislative success to those who are the most prominent on social media rather than the legislators who actually have influence.

For my second hypothesis considering legislator cosponsorship connectedness for the 115th and 116th Congress I will adjust my modeling approach. One issue researchers face when analyzing networks is endogeneity, which is caused when an unobservable individual characteristic influences both the link formation within the network and the outcome of interest. For example, it is plausible an omitted variable, such as social skills, is truly driving the underlying relationship between connectedness and legislative success, i.e. social skill is impacting the quantity and quality of connections within the network and social skill also increases legislative effectiveness. One way to address concerns of endogeneity is to include member fixed-effects in the model. I take this approach which controls for variation between congresspeople, essentially controlling for unobservable variables unique to each member of Congress.⁶

I also include a plethora of other alternative control variables that may impact legislative effectiveness originally laid out in Volden and Wiseman (2009). Specifically, controls are included for the length of time a member has been in Congress, the size of the delegation from the member state, the percent of the vote received to enter Congress, the first dimension of DW-Nominate corresponding to ideology (McCarty, Poole and Rosenthal, 1997), if the member is a committee chair, if the member is part of minority or

⁶Another way to deal with endogeneity is to implement a two-step Heckman correction process, following Battaglini, Sciabolazza and Patacchini (2020). The authors use alumni connections between legislators as an instrumental variable. However, the results indicate only the magnitude of the effect changes, not the sign or significance, so I do not perform this process.

	<i>Dependent variable:</i>
	Legislative Effectiveness Score
log(number of followers)	0.025 (0.040)
Seniority	0.016*** (0.006)
VoteShare	-0.007* (0.004)
DWNom	-0.039 (0.103)
SizeDel	0.002 (0.003)
ComChair	1.806*** (0.202)
PartyR	-0.671*** (0.116)
MajPLead	0.101 (0.352)
MinPLead	-0.065 (0.348)
Female	0.088 (0.107)
Speaker	-1.769* (0.915)
Black	-0.011 (0.143)
Latinx	-0.264 (0.168)
Constant	1.212*** (0.459)
Observations	406
R ²	0.382
Adjusted R ²	0.362
Residual Std. Error	0.827 (df = 392)
F Statistic	18.659*** (df = 13; 392)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Table 1.8: Regression output measuring the effect of the number of followers a legislator has on Twitter on legislative effectiveness for 116th House.

majority party leadership, gender, and controls for race/ethnicity. Lastly, for the House, a control is added for if they are Speaker of the House.

The output of the models for the cosponsorship and Twitter networks for both the House and Senate which include member fixed-effects is shown in Table 1.9. To use member fixed-effects between the 115th and 116th Congresses only the legislators who are present in both Congresses can be used, as a data point is needed for both time points. For the cosponsorship (Twitter) network, 341 (341) members of the U.S. House who were in the 115th House were also in the 116th House. Additionally, there were 91 (91) members from the 115th Senate returning for the 116th. The sample was then further reduced based on data available for the controls discussed above. The final sample for the House cosponsorship (Twitter) analysis contained 336 (333) legislators and the final sample for Senate cosponsorship contained 91 (91) members, as all Senators had complete data for the control variables. One can find regressions which do not use fixed-effects, and instead use all of the available legislators for the House and Senate in Appendix A. Importantly, the direction and statistical strength of the effect of connectedness on effectiveness did not change.

As one can see in 1.9, when including these controls as well as member fixed-effects the relationship between connectedness and legislative effectiveness is positive and significant at the .01 level. We interpret this as an increase in connectedness, either cosponsorship or Twitter connectedness, leading to an increase in legislative effectiveness. Put differently, more connected legislators are also more effective.

Connectedness was scaled to be between zero and one prior to this analysis, with one indicating high connectedness and zero indicating low connectedness. So, interpreting the results, going from the least connected congress person to the most connected congress person (a one-unit increase) would lead to a .262 increase in legislative effectiveness. Given legislator effectiveness score for the 116th Congress are between zero and ten, the substantive impact is small, but non negligible. However, due to the score being a combination of fifteen different measures of effectiveness, we can not tie this results

directly to a substantive interpretation, i.e. three more bills passed, or anything of this nature. Regardless, an increase in legislative effectiveness of .262 does indicate a legislator is becoming more productive at some aspect, or combination of aspects, of the indicators which make up the effectiveness score. Therefore, this effect should not be discounted. Additionally, for the Senate cosponsorship network, this value is .816, two times that of the House network, suggesting connectedness may play a bigger role in productivity within the Senate, possibly because it is easier to forge strong connections in the smaller legislative body.

If we turn to the results from the Twitter network, we see going from the least connectedness member of the House to the most connected member of the House (a one-unit increase) leads to a .316 increase in legislative effectiveness and a .733 increase within the Senate network. Once again demonstrating the importance of connectedness on legislative output. These results suggests Twitter is informing us about real connections within the House and Senate. They also inform us about legislative effectiveness as a by product of these connections, indicating the importance of social media for making or relaying legislative connections. Given the size standard errors, we can not make claims regarding the magnitude of the effect on legislative effectiveness between the cosponsorship and Twitter networks.

This analysis confirms Fowler (2006)'s findings and provides support for hypothesis 2. It also confirms the networks of retweets between legislators can lead to increased legislative effectiveness, providing support for hypothesis 3. This result is normatively impactful- as it demonstrates the network of retweets on Twitter allows insight into legislative output. I also ran an analysis which combined the House and Senate members who are legislators for both the 115th and 116th sessions of Congress. The results also demonstrate a positive and significant relationship between connectedness and effectiveness, the full output can be found in Appendix A.

Unfortunately, I can not run the same fixed effects research design looking at number of followers and connectedness within the same model because number of followers

is nested within the fixed effects. Therefore, I can not pit number of followers and connectedness against each other in a fixed effects model. However, for robustness' sake I provide a variety of supplementary analyses in Appendix A. Initially, in the most basic regression model it does appear the number of followers is related to legislative effectiveness. This result is highly sensitive and by adding other covariates the effect dissipates. Three models are run, a regression with the same catch-all covariates, the second is a model where only covariates which are doing work remain (seniority and Party), and the last is the parsimonious model discussed above only, which only includes connectedness and number of followers.

Additionally, one might suggest if I can not run a fixed effect model then I should instead run a random effects model. For completeness, random effect models with the same specifications as the three described above are run and the results are presented in Appendix A. Again, I find similar evidence that the number of followers a legislator has is not related to legislative effectiveness. Notably, in all six of these models the effect of connectedness on legislative effectiveness does not lose significance, providing evidence this result is robust to alternative model specifications.

	<i>Dependent variable:</i>			
	Legislative Effectiveness			
	Cosponsorship House	Cosponsorship Senate	Twitter House	Twitter Senate
Connectedness	0.262*** (0.068)	0.816*** (0.098)	0.316*** (0.067)	0.733*** (0.096)
Seniority	0.022 (0.065)	0.055 (0.040)	0.005 (0.065)	0.045 (0.040)
Size of Cong. Delegation	0.029 (0.078)	0.147 (0.126)	0.015 (0.078)	0.093 (0.126)
Vote Share Received 0.010	0.082 (0.070)	0.003 (0.061)	0.066 (0.070)	(0.061)
DW Nominate	0.007 (2.367)	−8.436* (4.665)	−0.257 (2.359)	−7.744 (4.676)
Party (I)	—	−3.263* (1.772)	—	−3.397* (1.774)
Party (R)	−0.633 (2.383)	7.328* (3.894)	−0.440 (2.375)	6.750* (3.904)
Committee Chair	−0.028 (1.736)	−3.154** (1.449)	−0.029 (1.730)	−2.749* (1.452)
Majority Party Leader	0.113 (0.711)	0.681 (1.224)	0.048 (0.709)	0.614 (1.227)
Minority Party Leader	−1.672 (4.053)	1.384 (0.924)	−1.151 (4.041)	1.626* (0.925)
Speaker	−1.116 (4.382)	—	0.112 (4.366)	—
Female	−0.125 (5.814)	−4.653* (2.591)	−0.235 (5.766)	−4.471* (2.596)
Black	−1.085 (4.748)	1.095 (0.964)	−0.650 (4.731)	1.106 (0.967)
Latinx	−1.085 (5.192)	1.095 (2.150)	−0.650 (5.174)	1.106 (2.154)
Constant	−0.467 (5.727)	−9.346 (5.652)	0.298 (5.708)	−7.676 (5.666)
Member Fixed-Effects	Yes	Yes	Yes	Yes
Observations	672	182	666	182
R ²	0.509	0.697	0.527	0.671
Adjusted R ²	0.017	0.391	0.053	0.338
Residual Std. Error	0.789 (df = 335)	0.515 (df = 90)	0.787 (df = 332)	0.516 (df = 90)
F Statistic	1.034 (df = 336; 335)	2.278*** (df = 91; 90)	1.112 (df = 333; 332)	2.016*** (df = 91; 90)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table 1.9: Fixed-Effects Output

1.7 Conclusion

Congress is a collaborative environment; members of Congress work with each other to advance legislation and further their congressional agendas. These collaborations signal underlying professional and social relationships. With the transition to the era of social media legislators have found new ways to keep their constituents informed of their work. However, these platforms also provide ways for legislators to interact and support each other's work. While prior research has focused on the relationships legislators have with constituents on these platforms (Barberá et al., 2013; Barberá et al., 2019; Russell, 2022) and how they use these platforms for policy framing (Hemphill, Russell and Schöpke-Gonzalez, 2021; Peterson et al., 2023), less attention has been paid to the interactions between legislators themselves. In this study, I focus on the connections between legislators, looking at their relationships in both the cosponsorship network and the network of Twitter interactions. I hypothesize these connections are indicative of legislative success. I argue this is because the relationships represent underlying social and professional connections that a legislator can use to advance legislation. However, I first claim this underlying network is not clear to everyday users on Twitter, and thus the impact legislators will have outside of Twitter is being misperceived.

A clear measure of influence on Twitter to the public is the number of followers a user has. Therefore, I first consider the relationship between the number of followers a legislator has and their effectiveness within Congress. The results indicate, in line with my first hypothesis, that the number of followers one has on Twitter is not related to legislative success. So, if an obvious measure of social influence cannot be tied to legislative success, what can Twitter tell us about legislator effectiveness, if anything? Is there a measure of influence from Twitter that does impact legislative success?

To explore this, I consider the legislative cosponsorship network and the legislative Twitter network for the 115th and 116th Congresses. I use Fowler's measure of connectedness to capture the extent of social relationships between legislators, the more connected a

legislator is, the more social influence they have with other legislators. I then run multiple models, controlling for the length of time a legislator has been in Congress as well as other controls previously identified in the literature as possible indicators of legislative success. My results indicate support for my second hypothesis, the more embedded a legislator is within the legislative cosponsorship network, the more effective they are as a legislator. As for my third hypothesis, which states the same relationship will hold for Twitter, the results indicate they do, even while controlling for endogeneity using member fixed-effects. This confirms interactions between legislators regardless of the environment they may be in (i.e. social media or formal legislative) signal legislative success.

My data, theory, and findings leave many open-ended questions to be explored in future research. First, the network of Twitter interactions between legislators is basically untapped. This study just scratched the surface of possible uses of this data. For example, one could look at levels of polarization, measured via cross-party ties, between the cosponsorship network and the legislative Twitter networks; is Twitter accurately representing polarization between legislators? What about legislative coalitions? Clearly, the results of this paper indicate Twitter interactions between legislators do have an impact outside of Twitter and warrant further study.

Additionally, the extension of the measure of connectedness into Twitter provides a theoretically more accurate measure of who is integral to a bill's process than cosponsorship. There are limitations to how many legislators one can tag in a tweet, but no limit to legislative cosponsorships. Therefore, one could explore how retweets correspond to legislative credit claiming. Future research could also attempt to quantify the impact the disconnect between influence on Twitter (measured via number of followers) and influence within Congress has on individuals' feelings towards the government. I expect this gap will lead to increased frustration.

One limitation of this study is the number of time points available. I currently utilized two sessions of Congress, leaving me with two observations. However, I could choose to split the sessions into year-long windows of time, giving four observations. I

could continue this and then split based on legislative sessions, i.e. split in half following the legislative break (eight observations) or continue splitting to some arbitrary number of observations. I could then collect the Twitter networks for the same shortened periods. I leave this as an area for future research.

Another limitation of this study is endogeneity. Specifically, legislators are strategic in who they cosponsor with and may be strategic in who they tweet and retweet on Twitter. This element of strategy makes it difficult to parse out the relationship between effectiveness and social connectedness as I may be missing a variable which may drive both legislative effectiveness and social connectedness (i.e. social skills). However, I utilized member fixed-effects to curb the effects of endogeneity.

My findings, along with the methods used, also have important implications beyond the scope of this study. For example, now that the measure of connectedness has been extended and validated using tweets instead of bills, it can be applied more broadly to research looking at the interactions between accounts on Twitter. While I was explicitly interested in studying the interactions between legislators on Twitter and measuring their influence, the method can just as easily be applied to study interactions between anyone on Twitter. The approach is also scalable to be used on millions of tweets, rather than the two million I considered. Substantively, my findings have important implications for the study of congressional effectiveness and its interplay with Twitter. Most importantly, I argue how influential one is within the legislative network impacts one's legislative effectiveness, and my results indicate that this relationship holds both within the legislative cosponsorship network and within the legislative Twitter network.

Finally, my results also demonstrate that not just any measure of social influence within Twitter will lead to increased legislative effectiveness. Particularly, there is not a relationship between the number of followers a legislative has and their effectiveness. Unfortunately, the number of followers a legislator has is the most public-facing measure of influence Twitter offers. Therefore, there might be a disconnect between which legislators Twitter expects to be effective and the actual effectiveness of that legislator. This should

thus encourage further research into the impact of this disconnect as well as thinking of other creative uses for the connectedness measure.

Bibliography

- 2021 Ballot Questions*. 2021. <https://vote.minneapolismn.gov/results-data/election-results/2021/ballot-questions/>.
- ABC Chicago*. 2018. <https://abc7chicago.com/van-dyke-trial-laquan-mcdonald-jason-case/4147086/>.
- Achen, Christopher H. and Larry M. Bartels. 2016. *Democracy for Realists: Why Elections Do Not Produce Responsive Government*. 1 ed. Princeton University Press.
URL: <http://www.jstor.org/stable/j.ctvc7770q>
- Alang, Sirry, Donna McAlpine and Malcolm McClain. 2021. “Police encounters as stressors: Associations with depression and anxiety across race.” *Socius* 7:2378023121998128.
- Anderson, William D, Janet M Box-Steffensmeier and Valeria Sinclair-Chapman. 2003. “The keys to legislative success in the US House of Representatives.” *Legislative Studies Quarterly* 28(3):357–386.
- Ang, Desmond and Jonathan Tebes. 2020. “Civic responses to police violence.”.
- Arnold, R Douglas and Nicholas Carnes. 2012. “Holding mayors accountable: New York’s executives from Koch to Bloomberg.” *American Journal of Political Science* 56(4):949–963.
- Barberá, Pablo, Richard Bonneau, John T Jost, Jonathan Nagler and Joshua Tucker. 2013. Is there anybody out there? The effects of legislators’ communication with their constituents. In *Text as Data Workshop*. London: London School of Economics and Political Science. pp. 1–24.
- Barberá, Pablo, Andreu Cases, Jonathan Nagler, Patrick J. Egan, Richard Bonneau, John T. Jost and Joshua A. Tucker. 2019. “Who Leads? Who Follows? Measuring Issue Attention and Agenda Setting by Legislators and the Mass Public Using Social Media Data.” *American Political Science Review* 113(4):883–901.
- Barrie, Christopher and Justin Ho. 2021. “academictwitterR: an R package to access the Twitter Academic Research Product Track v2 API endpoint.” *Journal of Open Source Software* 6(62):3272.
URL: <https://doi.org/10.21105/joss.03272>
- Battaglini, Marco, Valerio Leone Sciabolazza and Eleonora Patacchini. 2020. “Effectiveness of Connected Legislators.” *American Journal of Political Science* 64(4):739–756.
URL: <http://www.jstor.org/stable/45295346>

- Billingham, Chase M. 2019. "Within-District Racial Segregation and the Elusiveness of White Student Return to Urban Public Schools." *Urban Education* 54(2):151–181.
- Box-Steffensmeier, Janet M, Benjamin W Campbell, Andrew W Podob and Seth J Walker. 2020. "I get by with a little help from my friends: Leveraging campaign resources to maximize congressional power." *American Journal of Political Science* 64(4):1017–1033.
- Box-Steffensmeier, Janet M and Dino P Christenson. 2014. "The evolution and formation of amicus curiae networks." *Social Networks* 36:82–96.
- Box-Steffensmeier, Janet M, Dino P Christenson and Alison W Craig. 2019. "Cue-taking in congress: Interest group signals from dear colleague letters." *American Journal of Political Science* 63(1):163–180.
- Box-Steffensmeier, Janet M, Dino P Christenson and Matthew P Hitt. 2013. "Quality over quantity: Amici influence and judicial decision making." *American Political Science Review* 107(3):446–460.
- Bratton, Kathleen A and Kerry L Haynie. 1999. "Agenda setting and legislative success in state legislatures: The effects of gender and race." *The Journal of Politics* 61(3):658–679.
- Brown, Jacob R and Michael Zoorob. 2022. "Resisting Broken Windows." *Political Behavior* 44(2):679–703.
- Brown v. Board of Education of Topeka. 1954. *U.S.* 347(483).
- Bunyasi, Tehama Lopez and Candis Watts Smith. 2019. *Stay Woke: A People's Guide to Making All Black Lives Matter*.
- Burkett, Tracy Lynn. 1997. *Cosponsorship in the United States Senate: A network analysis of Senate communication and leadership, 1973-1990*. University of South Carolina.
- Burnett, Craig M and Vladimir Kogan. 2017. "The politics of potholes: Service quality and retrospective voting in local elections." *The Journal of Politics* 79(1):302–314.
- Butler, Daniel M., Adam G. Hughes, Craig Volden and Alan E. Wiseman. 2021. "Do constituents know (or care) about the lawmaking effectiveness of their representatives?" *Political Science Research and Methods* p. 1–10.
- Caldeira, Gregory A and Samuel C Patterson. 1987. "Political friendship in the legislature." *The Journal of Politics* 49(4):953–975.
- Caldera, Sophia, Daryl DeFord, Moon Duchin, Samuel C. Gutekunst and Cara Nix. 2020. "Mathematics of Nested Districts: The Case of Alaska." *Statistics and Public Policy* 7(1):39–51.
URL: <https://doi.org/10.1080/2330443X.2020.1774452>
- Cha, Meeyoung, Hamed Haddadi, Fabricio Benevenuto and Krishna Gummadi. 2010. Measuring user influence in twitter: The million follower fallacy. In *Proceedings of the international AAAI conference on web and social media*. Vol. 4 pp. 10–17.

- Chatterji, Roby. 2020. "Fighting Systemic Racism in K-12 Education: Helping Allies Move From the Keyboard to the School Board." *Center for American Progress* .
URL: <https://www.americanprogress.org/issues/education-k-12/news/2020/07/08/487386/fighting-systemic-racism-k-12-education-helping-allies-move-keyboard-school-board/>
- Chen, Jowei, Jonathan Rodden et al. 2013. "Unintentional gerrymandering: Political geography and electoral bias in legislatures." *Quarterly Journal of Political Science* 8(3):239–269.
- Cho, Wendy K.Tam and Yan Y. Liu. 2021. "A parallel evolutionary multiple-try metropolis Markov chain Monte Carlo algorithm for sampling spatial partitions." *Statistics and Computing* 31(1):10.
URL: <http://link.springer.com/10.1007/s11222-020-09977-z>
- Civettini, Andrew JW and David P Redlawsk. 2009. "Voters, emotions, and memory." *Political Psychology* 30(1):125–151.
- Cohen, Lauren and Christopher J Malloy. 2014. "Friends in high places." *American Economic Journal: Economic Policy* 6(3):63–91.
- Coleman, James S., Sara D. Kelly and John A. Moore. 1975. "Trends in School Segregation 1968-73." *The Urban Institute* .
- Craig, Alison W. 2021. "It Takes a Coalition: The Community Impacts of Collaboration." *Legislative Studies Quarterly* 46(1):11–48.
- Crain, Robert L. and Rita E. Mahard. 1978. "School Racial Composition and Black College Attendance and Achievement Test Performance." *Sociology of Education* 51(2):81.
URL: </record/1979-14411-001>
- Darling-Hammond, Linda. 1998. "Unequal Opportunity: Race and Education." *The Brookings Review* 16(2):28.
- Darling-Hammond, Linda. 2004. "STATE OF THE DISCIPLINE THE COLOR LINE IN Race , Resources , and Student Achievement." *Du Bois Review* 1(2):213–246.
- Darling-Hammond, Linda, Brian D. Smedley, Adrienne Y. Stith, Lois Colburn, Clyde H. Evans and Institute of Medicine (US). 2001. "Inequality in Teaching and Schooling: How Opportunity Is Rationed to Students of Color in America." *National Academies Press (US)* .
URL: <https://www.ncbi.nlm.nih.gov/books/NBK223640/>
- DeFord, Daryl, Moon Duchin and Justin Solomon. 2021. "Recombination: A family of Markov chains for redistricting." *Harvard Data Science Review* .
- Desmond, Matthew, Andrew V Papachristos and David S Kirk. 2016. "Police violence and citizen crime reporting in the black community." *American sociological review* 81(5):857–876.

- Dijkstra, Edsger W. 1959. "A note on two problems in connexion with graphs." *Numerische mathematik* 1(1):269–271.
- Drakulich, Kevin, Eric Rodriguez-Whitney and Jesenia Robles. 2023. "Why White Americans more frequently fail to view the police critically: A subtle but vital shift in focus." *Du Bois Review: Social Science Research on Race* 20(1):57–88.
- EdChoice. 2024.
URL: <https://edchoice.infogram.com/edchoice-share-2024-1h9j6gg3evmdv4g>
- Fiel, Jeremy E. 2013. "Decomposing School Resegregation: Social Closure, Racial Imbalance, and Racial Isolation." *American Sociological Review* 78(5):828–848.
- Fiorina, Morris P. 1976. "The voting decision: instrumental and expressive aspects." *The journal of politics* 38(2):390–413.
- Fiorina, Morris P. 1978. "Economic retrospective voting in American national elections: A micro-analysis." *American Journal of political science* pp. 426–443.
- Fowler, James H. 2006. "Connecting the congress: A study of cosponsorship networks." *Political Analysis* 14(4):456–487.
- Garcia-Montoya, Laura, Ana Arjona and Matthew Lacombe. 2022. "Violence and voting in the United States: How school shootings affect elections." *American Political Science Review* 116(3):807–826.
- Gerber, Alan S, James G Gimpel, Donald P Green and Daron R Shaw. 2011. "How large and long-lasting are the persuasive effects of televised campaign ads? Results from a randomized field experiment." *American Political Science Review* 105(1):135–150.
- Gill v. Whitford. 2018. *U.S.* (585).
- Go, Min Hee. 2022. "Holding mayors accountable? Policing and mayoral approval in American cities." *The Social Science Journal* pp. 1–16.
- Golbeck, Jennifer, Justin M. Grimes and Anthony Rogers. 2010. "Twitter use by the U.S. Congress." *Journal of the American Society for Information Science and Technology* 61(8):1612–1621.
URL: <https://onlinelibrary.wiley.com/doi/full/10.1002/asi.21344><https://onlinelibrary.wiley.com/doi/abs/10.1002/asi.21344><https://onlinelibrary.wiley.com/doi/10.1002/asi.21344>
- Güss, C Dominik, Ma Teresa Tuason and Alicia Devine. 2020. "Problems with police reports as data sources: a researchers' perspective." *Frontiers in Psychology* 11:582428.
- Hajnal, Zoltan and Jessica Trounstein. 2014. "Identifying and understanding perceived inequities in local politics." *Political Research Quarterly* 67(1):56–70.
- Hajnal, Zoltan L and Jessica Trounstein. 2010. "Who or what governs?: The effects of economics, politics, institutions, and needs on local spending." *American Politics Research* 38(6):1130–1163.

- Hajnal, Zoltan L, Vladimir Kogan and G Agustin Markarian. 2022. “Who Votes: City Election Timing and Voter Composition.” *American Political Science Review* 116(1):374–383.
- Hanushek, Eric A, John F Kain and Steven G Rivkin. 2000. “How Much Does School Integration Affect Student Achievement?” *Association for Public Policy Analysis and Management*, 2000 .
- Hawkins. 2010. “Eden Prairie School Board to Grapple with Contentious Boundary Plan Tonight.”
URL: <https://www.minnpost.com/learning-curve/2010/11/eden-prairie-school-board-grapple-contentious-boundary-plan-tonight/>
- Healy, Andrew and Neil Malhotra. 2013. “Retrospective voting reconsidered.” *Annual review of political science* 16:285–306.
- Hemphill, Libby, Annelise Russell and Angela M Schöpke-Gonzalez. 2021. “What drives US congressional members’ policy attention on Twitter?” *Policy & Internet* 13(2):233–256.
- Hopkins, Daniel J and Lindsay M Pettingill. 2018. “Retrospective voting in big-city US mayoral elections.” *Political Science Research and Methods* 6(4):697–714.
- Ingraham, Christopher. 2020. “Racial Inequality in Minneapolis Is among the Worst in the Nation.” *The Washington Post* .
URL: www.washingtonpost.com/business/2020/05/30/minneapolis-racial-inequality
- ISD 622. 2021. “Boundary Project / School Boundary Committee Charge and Guiding Principles.”
URL: <https://www.isd622.org/Page/15252>
- Jackson, C Kirabo, Rucker C Johnson and Claudia Persico. 2015. The effects of school spending on educational and economic outcomes: Evidence from school finance reforms. Technical report National Bureau of Economic Research.
- Johnson, Rucker C. and Alexander Nazaryan. 2019. *Children of the dream : why school integration works*.
- Key, Valdimer Orlando. 1966. *The responsible electorate: Rationality in presidential voting 1936–1960*. Harvard University Press.
- Kirkland, Justin H. 2011. “The relational determinants of legislative outcomes: Strong and weak ties between legislators.” *The Journal of Politics* 73(3):887–898.
- Kirkland, Justin H. 2013. “Hypothesis testing for group structure in legislative networks.” *State Politics & Policy Quarterly* 13(2):225–243.
- Kirkland, Justin H and Justin H Gross. 2014. “Measurement and theory in legislative networks: The evolving topology of Congressional collaboration.” *Social networks* 36:97–109.

- Klein Teeselink, Bouke and Georgios Melios. 2021. "Weather to Protest: The Effect of Black Lives Matter Protests on the 2020 Presidential Election." *Available at SSRN 3809877*.
- Kramer, Gerald H. 1971. "Short-term fluctuations in US voting behavior, 1896–1964." *American political science review* 65(1):131–143.
- Lallinger, Stefan. 2023. "Is the Fight for School Integration Still Worthwhile for African Americans?"
URL: <https://tcf.org/content/report/is-the-fight-for-school-integration-still-worthwhile-for-african-americans/>
- LaPira, Timothy M, Herschel F Thomas and Frank R Baumgartner. 2009. The structure and stability of lobbying networks in washington. In *annual meeting of the Midwest Political Science Association*. Citeseer.
- Lerman, Amy E and Vesla Weaver. 2014. "Staying out of sight? Concentrated policing and local political action." *The ANNALS of the American Academy of Political and Social Science* 651(1):202–219.
- Libassi, CJ. 2018. "The Neglected College Race Gap_ Racial Disparities Among College Completers - Center for American Progress."
URL: <https://www.americanprogress.org/issues/education-postsecondary/reports/2018/05/23/451186/neglected-college-race-gap-racial-disparities-among-college-completers/>
- Liebertson, Stanley. 1969. "Measuring Population Diversity." *American Sociological Review* 34(6):850–62.
- Logan, John R., Deirdre Oakley and Jacob Stowell. 2006. "Resegregation in U.S. Public Schools or White Decline? A Closer Look at Trends in the 1990s." *Children, Youth and Environments* 16(1):49–68.
URL: <https://www.journals.uchicago.edu/doi/abs/10.1086/587150>
- Masket, Seth E. 2008. "Where you sit is where you stand: The impact of seating proximity on legislative cue-taking." *Quarterly Journal of Political Science* 3:301–311.
- Massey, Douglas S . and Nancy A . Denton. 1988. "The Dimensions of Residential Segregation Author (s): Douglas S . Massey and Nancy A . Denton Published by : University of North Carolina Press Stable URL : <http://www.jstor.org/stable/2579183>." *Social Forces* 67(2):281–315.
- maup. 2019. "MAUP: The geospatial toolkit for redistricting data." <https://github.com/mggg/maup>.
- McCartan, Cory and Kosuke Imai. 2023. "Sequential Monte Carlo for sampling balanced and compact redistricting plans." *The Annals of Applied Statistics* 17(4):3300–3323.
- McCarty, Nolan, Keith T Poole and Howard Rosenthal. 1997. "Income Redistribution and the Realignment of American Politics (Washington, DC: American Enterprise Institute Press).".

- Mergel, Ines. 2012. *Social Media in the Public Sector: Participation, Collaboration, and Transparency in the Networked World*.
- MGGG. 2017. "GerryChain." <https://github.com/mggg/GerryChain>.
- Miller, Michael K. 2013. "For the win! The effect of professional sports records on mayoral elections." *Social Science Quarterly* 94(1):59–78.
- Milliken v. Bradley. 1974. *U.S.* 418(717).
- Minnesota Department of Education. 2020. "Open Enrollment." .
URL: <https://education.mn.gov/mde/fam/open/>
- Minnesota Department of Employment and Economic Development. 2022. "Black Minnesotans: An analysis of employment, educational and other economic trends and disparities related to the labor force." .
URL: <https://mn.gov/deed/newscenter/publications/trends/september-2022/disparities.jsp>
- MN Geospatial Commons. 2019. "School Attendance Area Boundaries, Minnesota, SY2019-2020." .
URL: <https://gisdata.mn.gov/dataset/bdry-school-attendance-areas>
- MN Rule 3535.0100. 1999.
URL: <https://www.revisor.mn.gov/rules/3535.0100/>
- MN Rule 3535.0110. 1999.
URL: <https://www.revisor.mn.gov/rules/3535.0100/>
- NAACP. 2019. "Economic Inclusion Plan." .
URL: <https://www.naacp.org/wp-content/uploads/2019/12/EconomicInclusionPlanTwinCities-3.pdf>
- National Center for Educational Statistics. 2018. "Public High School Graduation Rates." *The Condition of Education* pp. 1–7.
URL: <https://www.alsde.edu/sec/comm/NewsReleases/12-08-2016GraduationRateReview.pdf>.[https://nces.ed.gov/programs/coe/indicator{_-}coi.asp](https://nces.ed.gov/programs/coe/indicator/coi.asp)
- NCES. 2016. "School Attendance Boundary Collection." *National Center for Education Statistics* .
URL: <https://nces.ed.gov/programs/edge/SABS>
- NCES. 2019. "School District Geographic Relationship Files." *National Center for Education Statistics* .
URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=1647221.
- Nuamah, Sally A and Thomas Ogorzalek. 2021. "Close to home: Place-based mobilization in racialized contexts." *American Political Science Review* 115(3):757–774.

- Oliver, J Eric and Shang E Ha. 2007. "Vote choice in suburban elections." *American Political Science Review* 101(3):393–408.
- Orfield, Gary. 2009. "Reviving the Goal of an Integrated Society: A 21st Century Challenge."
- Peng, Tai-Quan, Mengchen Liu, Yingcai Wu and Shixia Liu. 2016. "Follower-followee network, communication networks, and vote agreement of the US members of congress." *Communication research* 43(7):996–1024.
- Penner, Andrew M. 2008. "Race and gender differences in wages: The role of occupational sorting at the point of hire." *Sociological Quarterly* 49(3):597–614.
- Perez. 2016. <https://www.chicagotribune.com/politics/ct-rahm-emanuel-laquan-mcdonald-poll-20160131-story.html>.
- Peterson, David AM, Wallapak Tavanapong, Lei Qi, Adisak Sukul and Mohammed Khaleel. 2023. "The public-facing policy agenda of state legislatures: The communication of public policy via twitter." *Policy Studies Journal* 51(3):551–571.
- Pew Research. 2020. <https://www.pewresearch.org/social-trends/2020/06/12/amid-protests-majorities-across-racial-and-ethnic-groups-express-support-for-the-black-lives-matter-movement/>.
- Porter, Mason A, Peter J Mucha, Mark EJ Newman and Casey M Warmbrand. 2005. "A network analysis of committees in the US House of Representatives." *Proceedings of the National Academy of Sciences* 102(20):7057–7062.
- Reardon, Sean F. and Ann Owens. 2014. "60 years after brown: Trends and consequences of school segregation." *Annual Review of Sociology* 40:199–218.
- Reardon, Sean F., John T. Yun and Tamela McNulty Eitle. 2000. "The changing structure of school segregation: Measurement and evidence of multiracial metropolitan-area school segregation, 1989-1995." *Demography* 37(3):351–364.
- Rich, Steven, Andrew Ba Tran and Jennifer Jenkins. 2022. "Fatal police shootings are still going up, and nobody knows why." <https://www.washingtonpost.com/investigations/2023/02/21/fatal-police-shootings-increase-2022/>.
- Rocca, Michael S and Gabriel R Sanchez. 2008. "The effect of race and ethnicity on bill sponsorship and cosponsorship in Congress." *American Politics Research* 36(1):130–152.
- Russell, Annelise. 2022. "Constituent connections: senators' reputation building in the age of social media." *Journal of Information Technology & Politics* 19(2):180–196.
- Samuels A., Christina. 2011. "School Boundary Plan Divides Minn. Suburb."
URL: <https://www.edweek.org/leadership/school-boundary-plan-divides-minn-suburb/2011/10>

- Shearer, Elisa. 2021. “More than eight-in-ten Americans get news from digital devices.”
URL: <https://www.pewresearch.org/fact-tank/2021/01/12/more-than-eight-in-ten-americans-get-news-from-digital-device>
- Shoub, Kelsey and Kevin Morris. 2022. “Contested Killings: The Mobilizing Effects of Community Contact with Police Violence.”.
- Sosina, Victoria E. and Ericka S. Weathers. 2019. “Pathways to Inequality: Between-District Segregation and Racial Disparities in School District Expenditures.” *AERA Open* 5(3):233285841987244.
- Turner, Margery Austin, Matthew M Chingos and Natalie Spievack. 2021. “White People’s Choices Perpetuate School and Neighborhood Segregation What Would It Take to Change Them?”.
URL: <https://educationdata.urban.org/documentation/>,
- Twitter API for Academic Research. 2021. <https://developer.twitter.com/en/products/twitter-api/academic-research>.
- U.S. Census Bureau. 2019. “American Community Survey MEDIAN HOUSEHOLD INCOME IN THE PAST 12 MONTHS (IN 2019 INFLATION-ADJUSTED DOLLARS).”
URL: <https://data.census.gov/cedsci/table?t=Income%20and%20Poverty&tid=ACSDT5Y2019.B19013>
- US Department of Education. 2017. “State Performance Reports.”
URL: <https://oese.ed.gov/offices/office-of-administration/about-us/consolidated-state-performance-reports/#sy15-16>
- Volden, Craig and Alan E. Wiseman. 2009. “Legislative Effectiveness in Congress.”.
- Volden, Craig, Alan E Wiseman and Dana E Wittmer. 2013. “When are women more effective lawmakers than men?” *American Journal of Political Science* 57(2):326–341.
- Warshaw, Christopher, Justin de Benedictis-Kessner and Yamil Velez. 2022. “Local Representation in the United States: A New Comprehensive Dataset of Elections.”.
- Washington Post Fatal Force Database. 2023. <https://www.washingtonpost.com/graphics/investigations/police-shootings-database/>.
- Wasow, Omar. 2020. “Agenda seeding: How 1960s black protests moved elites, public opinion and voting.” *American Political Science Review* 114(3):638–659.
- Weaver, Vesla M and Amy E Lerman. 2010. “Political consequences of the carceral state.” *American Political Science Review* 104(4):817–833.
- Weitzer, Ronald. 2002. “Incidents of police misconduct and public opinion.” *Journal of criminal justice* 30(5):397–408.
- White, Ariel. 2019. “Misdemeanor disenfranchisement? The demobilizing effects of brief jail spells on potential voters.” *American Political Science Review* 113(2):311–324.

- Whitler, Melissa. 2023. "How Minnesota's school funding leaves the most in-need districts behind." **URL:** <https://www.mplsschoolsvoices.news/posts/mn-school-funding-inequities-data-viz>
- Wilson, Rick K. and Cheryl D. Young. 1997. "Cosponsorship in the U. S. Congress." *Legislative Studies Quarterly* 22(1):25.
- Wojcik, Stefan and Adam Hughes. 2019. "Sizing Up Twitter Users." **URL:** <https://www.pewresearch.org/internet/2019/04/24/sizing-up-twitter-users/>
- Zoorob, Michael. 2020. "Do police brutality stories reduce 911 calls? Reassessing an important criminological finding." *American sociological review* 85(1):176–183.

Appendix A: Social Influence and its Translation to the Legislative Arena

The appendix contains the following information:

1. Connectedness Tables;
2. Connectedness Linear Regressions;
3. Extended Analyses;
4. Robustness Analyses.

A.1 Connectedness Tables

When considering who are the most connected within each contextual environment there is no overlap between the top five legislators for the 116th Senate as can be seen in Tables A.1 and A.2. No, or very little, overlap also occurs for both the 115th House and 115th Senate. For brevity, those tables are not included.

A.2 Connectedness Linear Regressions

Because fixed-effects regressions require an individual to have been in both the 115th and 116th Congresses, I also ran separate regressions which did not include fixed

Legislator	Connectedness
Sen. Blumenthal, Richard [D-CT]	1.49
Sen. Brown, Sherrod [D-OH]	1.29
Sen. Markey, Edward J. [D-MA]	1.28
Sen. Van Hollen, Chris [D-MD]	1.28
Sen. Warren, Elizabeth [D-MA]	1.27

Table A.1: Connectedness: Top five legislators within the 116th Senate cosponsorship network.

Legislator	Connectedness
Sen. Schumer, Charles E. [D-NY]	4.93
Sen. Harris, Kamala D. [D-CA]	4.69
Sen. Markey, Edward J. [D-MA]	4.55
Sen. Booker, Cory A. [D-NJ]	4.45
Sen. Sanders, Bernard [I-VT]	4.40

Table A.2: Connectedness: Top five legislators within the 116th Senate Twitter network.

effects. I find a positive and significant relationship between cosponsorship connectedness and legislative effectiveness for both chambers of Congress and for both sessions of Congress considered in this analysis, see Table A.3. I also find a positive and significant relationship between Twitter connectedness and legislative effectiveness for the 116th House, 115th, and the 115th Senate, see Table A.4. This indicates those who are the most connected within the Twitter network are also more effective legislators. Notably, there are differences in both the magnitude of the coefficients as well as the level of significance between Congresses.

Though Twitter launched in 2006, legislators were relatively slow to adopt the platform and understand how to use it to their advantage. I believe the varying significance found in Table A.4 between the 115th and 116th Congresses is partly because the relationship between Twitter connectedness and legislative effectiveness became stronger as legislators started to realize how valuable the platform could be for them and how to use

	116 House Estimate (S.E.)	116 Senate Estimate (S.E.)	115 House Estimate (S.E.)	115 Senate Estimate (S.E.)
(Intercept)	-1.111** (0.413)	0.003 (0.563)	0.289 (0.321)	-0.344 (0.546)
Connectedness	9.406*** (2.235)	0.950 . (0.508)	3.933* (1.934)	1.782* (0.788)
Seniority	0.037*** (0.005)	0.003 (0.010)	0.006 (0.005)	0.015 . (0.008)
N	444	101	443	104
RMSE	0.962	0.809	0.907	0.751
R^2	0.129	0.035	0.013	0.066
adj R^2	0.125	0.015	0.009	0.047
$.p \leq 0.1$ $*p \leq 0.05$ $**p \leq 0.01$ $***p \leq 0.001$				

Table A.3: Regression output for measuring the effect of connectedness within the cosponsorship network on legislative effectiveness.

the platform to their advantage following the election of Rep. Alexandria Ocasio-Cortez. Therefore, the relationship between connectedness and effectiveness may not have existed in earlier congresses than the ones studied here and became a powerful force beginning with the 116th Congress.

A.3 Extended Analyses

The results for a combined analysis with the House and Senate are included in Table A.5. Member fixed-effects are included as well as the controls discussed earlier. The relationship between connectedness and legislative effectiveness still holds and is positive and significant. This indicates for the legislative body as a whole, those who are more connected on Twitter are also more effective legislators.

	116 House Estimate (S.E.)	116 Senate Estimate (S.E.)	115 House Estimate (S.E.)	115 Senate Estimate (S.E.)
(Intercept)	-0.071 (0.192)	0.837* (0.390)	0.621*** (0.155)	0.415 (0.259)
Connectedness	0.358*** (0.094)	0.057 (0.111)	0.394* (0.180)	0.407 (0.208)
Seniority	0.034*** (0.005)	-0.002 (0.009)	0.006 (0.005)	0.009 (0.008)
N	435	101	443	104
RMSE	0.966	0.823	0.907	0.756
R^2	0.120	0.003	0.015	0.054
adj R^2	0.116	-0.017	0.010	0.036

$.p \leq 0.1$ * $p \leq 0.05$ ** $p \leq 0.01$ *** $p \leq 0.001$

Table A.4: Regression output for measuring the effect of connectedness within Twitter on legislative effectiveness.

	<i>Dependent variable:</i>	
	Legislative Effectiveness	
	Cosponsorship	Twitter
Connectedness	0.356*** (0.059)	0.387*** (0.057)
Seniority	0.042 (0.056)	0.036 (0.056)
Party (I)	−3.538 (2.408)	−3.440 (2.385)
Party (R)	9.134 (5.733)	8.160 (5.681)
VoteShare	0.018 (0.042)	0.017 (0.042)
DWNom	−10.878 (6.995)	−9.660 (6.933)
Size of Delegation	0.058 (0.174)	0.026 (0.173)
Committee Chair	−2.149 (1.839)	−1.800 (1.822)
Majority Party	0.699 (1.812)	0.483 (1.795)
Minority Party	1.443 (1.276)	1.520 (1.264)
Female	−5.819 (3.904)	−5.268 (3.868)
Black	1.790 (1.387)	1.702 (1.375)
Latinx	1.081 (3.058)	1.609 (3.030)
Constant	−5.372 (6.154)	−4.625 (6.080)
Member Fixed Effects	Yes	Yes
Observations	854	848
R ²	0.525	0.540
Adjusted R ²	0.049	0.079
Residual Std. Error	0.750 (df = 426)	0.743 (df = 423)
F Statistic	1.103 (df = 427; 426)	1.170* (df = 424; 423)
<i>Note:</i>		*p<0.1; **p<0.05; ***p<0.01

Table A.5: Combined House and Senate regression with member fixed effects.

A.4 Robustness Analyses

Two different modeling techniques were completed: regression and random effects. Table A.6 presents three regressions were run with varying numbers of covariates, first, a parsimonious model looking at legislator connectedness within the cosponsorship network and the log number of followers a legislator has and their impact on legislative effectiveness. As one can see, initially it seems as if the number of followers a legislator has does impact legislative effectiveness. However, these results are highly sensitive, and the inclusion of more control variables, even just the two main covariates which are doing work (Party and Seniority) is enough to strip away the significance from the number of followers. Notably, connectedness does not share this sensitivity and the results are robust to these alternative model specifications. These results are also mirrored if we consider a legislator's connectedness on Twitter, as seen in Table A.7.

Identical patterns are also displayed in Tables A.8 and A.9 in which random effect models with the same specifications were run. Recall I could not run a fixed effect model as I would have preferred due to the number of followers being nested within the member fixed effects. Taken together, this indicates the number of followers a legislator have does not impact a legislator's legislative effectiveness score, only in the most naive parsimonious models do we find this effect, and the inclusion of other controlling covariates demonstrates they are doing the real work, not the number of followers.

<i>Dependent variable:</i>			
	Legislative Effectiveness		
	(1)	(2)	(3)
Cosponsorship Connectedness	0.300*** (0.053)	0.423*** (0.051)	0.427*** (0.053)
log(number of followers)	0.080*** (0.030)	0.016 (0.028)	0.033 (0.031)
Seniority		0.013*** (0.004)	0.015*** (0.004)
PartyR		−0.552*** (0.063)	−0.527** (0.223)
VoteShare			−0.00000 (0.002)
DWNom			−0.026 (0.252)
SizeDel			0.0004 (0.002)
ComChair			−0.170 (0.170)
MajPLead			−0.052 (0.221)
MinPLead			−0.195 (0.267)
Female			0.102 (0.080)
Speaker			−1.321** (0.594)
Black			−0.098 (0.099)
Latinx			−0.166 (0.128)
Constant	−0.235 (0.318)	0.416 (0.302)	0.209 (0.343)
Observations	632	632	632
R ²	0.057	0.198	0.214
Adjusted R ²	0.054	0.193	0.196
Residual Std. Error	0.789 (df = 629)	0.729 (df = 627)	0.727 (df = 617)
F Statistic	18.983*** (df = 2; 629)	38.798*** (df = 4; 627)	12.006*** (df = 14; 617)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table A.6: Results of regression analysis with various specifications looking at connectedness within the cosponsorship network the number of followers a legislator has.

	<i>Dependent variable:</i>		
	Legislative Effectiveness		
	(1)	(2)	(3)
Twitter Connectedness	0.292*** (0.052)	0.427*** (0.050)	0.434*** (0.053)
log(number of followers)	0.107*** (0.030)	0.040 (0.028)	0.048 (0.031)
Seniority		0.014*** (0.004)	0.015*** (0.004)
PartyR		−0.560*** (0.063)	−0.540** (0.222)
VoteShare			0.001 (0.002)
DWNom			−0.025 (0.251)
SizeDel			0.0001 (0.002)
ComChair			−0.196 (0.171)
MajPLead			−0.073 (0.221)
MinPLead			−0.075 (0.267)
Female			0.115 (0.080)
Speaker			−0.966 (0.593)
Black			−0.130 (0.099)
Latinx			−0.172 (0.127)
Constant	−0.522* (0.316)	0.143 (0.300)	−0.025 (0.342)
Observations	632	632	632
R ²	0.066	0.211	0.223
Adjusted R ²	0.063	0.206	0.206
Residual Std. Error	0.788 (df = 629)	0.725 (df = 627)	0.725 (df = 617)
F Statistic	22.047*** (df = 2; 629)	41.954*** (df = 4; 627)	12.662*** (df = 14; 617)

Note:

*p<0.1; **p<0.05; ***p<0.01

Table A.7: Results of regression analysis with various specifications looking at connectedness within the Twitter network the number of followers a legislator has.

	<i>Dependent variable:</i>		
	Legislative Effectiveness		
	(1)	(2)	(3)
Cosponsorship Connectedness	0.300*** (0.053)	0.423*** (0.051)	0.427*** (0.053)
log(number of followers)	0.080*** (0.030)	0.016 (0.028)	0.033 (0.031)
Seniority		0.013*** (0.004)	0.015*** (0.004)
PartyR		-0.552*** (0.063)	-0.527** (0.223)
VoteShare			-0.00000 (0.002)
DWNom			-0.026 (0.252)
SizeDel			0.0004 (0.002)
ComChair			-0.170 (0.170)
MajPLead			-0.052 (0.221)
MinPLead			-0.195 (0.267)
Female			0.102 (0.080)
Speaker			-1.321** (0.594)
Black			-0.098 (0.099)
Latinx			-0.166 (0.128)
Constant	-0.235 (0.318)	0.416 (0.302)	0.209 (0.343)
Observations	632	632	632
Log Likelihood	-752.599	-708.057	-718.676
Akaike Inf. Crit.	1,515.199	1,430.113	1,471.351
Bayesian Inf. Crit.	1,537.443	1,461.255	1,546.982

Note:

*p<0.1; **p<0.05; ***p<0.01

Table A.8: Results of random effects analysis with various specifications looking at connectedness within the cosponsorship network the number of followers a legislator has.

	<i>Dependent variable:</i>		
	Legislative Effectiveness		
	(1)	(2)	(3)
Twitter Connectedness	0.292*** (0.052)	0.427*** (0.050)	0.434*** (0.053)
log(number of followers)	0.107*** (0.030)	0.040 (0.028)	0.048 (0.031)
Seniority		0.014*** (0.004)	0.015*** (0.004)
PartyR		-0.560*** (0.063)	-0.540** (0.222)
VoteShare			0.001 (0.002)
DWNom			-0.025 (0.251)
SizeDel			0.0001 (0.002)
ComChair			-0.196 (0.171)
MajPLead			-0.073 (0.221)
MinPLead			-0.075 (0.267)
Female			0.115 (0.080)
Speaker			-0.966 (0.593)
Black			-0.130 (0.099)
Latinx			-0.172 (0.127)
Constant	-0.522* (0.316)	0.143 (0.300)	-0.025 (0.342)
Observations	632	632	632
Log Likelihood	-751.859	-705.160	-717.171
Akaike Inf. Crit.	1,513.719	1,424.321	1,468.342
Bayesian Inf. Crit.	1,535.963	1,455.463	1,543.973

Note: *p<0.1; **p<0.05; ***p<0.01

Table A.9: Results of random effects analysis with various specifications looking at connectedness within the Twitter network the number of followers a legislator has.